MILITARY SPECIFICATION

TEST SET, ELECTRONIC SYSTEM: M137

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, and preparation for delivery for Test Set, Electronics System: M137.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-28800	-	Test Equipment for Use With Electronic Equipment, General Specification for.
MIL-W-63150	-	Weapons and Support Material, Standard Quality Assurance Provision for.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-280	-	Definitions of Item Levels, Item Exchangeability, Models and Related Terms.

FSC 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-1235 - Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein:

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT

11831041 - Test Set, Electronics System.

PACKAGING

P11831041 - Test Set, Electronics System.

PUBLICATIONS

HEADQUARTERS, DEPARTMENT OF THE ARMY

Technical Manual

TM 9-4933-244-13 & P Aviation Unit and Intermediate Maintenance Instructions (Including Repair Parts and Special Tool Lists) for Electronics System Test Set: M137.

(Copies of specifications, standards, handbooks, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Item definition. The M137 Electronics System Test Set, Drawing 11831041, shall be used to verify proper operation of the Armament Subsystem Helicopter, 20MM Automatic Gun: M97E1 and M97E2, which includes the components specified in Table I. The test set consists of two inclosed military-type portable cases. One case contains all the required circuitry, switches, and indicators; the other case contains all the interconnecting cabling necessary to interface the test set to the armament subsystem components as well as the self-test cable and the firing simulator assembly. The test set shall provide the operator with the ability to perform a functional checkout of the armament sybsystem and fault diagnosis to a defective line replaceable unit (LRU) and further fault isolation of that LRU to a shop replaceable unit (SRU). Additionally, the test set provides adjustment and alignment capability of system components.

3.1.1 Government furnished property list. Each test set shall include a government furnished technical manual, TM9-4933-224-13 & P.

3.2 Materials. Materials shall be in accordance with applicable drawings, specifications and standards.

3.3 Assembly. The M137 Electronics System Test Set assembly shall comply with all requirements specified on Drawing (Dwg) 11831041, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.4 Interchangeability. All test sets manufactured to this specification shall be functionally interchangeable as defined in MIL-STD-280. Interchangeability shall exist between units and all replaceable assemblies, subassemblies and parts having the same part number.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article inspection by the contractor shall be as specified in the contract and 4.4.

3.6 Performance characteristics.

3.6.1 Performance. The test set shall be capable of testing the Armament Subsystem, Helicopter, 20MM Automatic Gun: M97E1 and M97E2, as described in 3.1, and shall have the performance characteristics specified herein.

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3.6.1.1 Self-test.

3.6.1.1.1 Initialization. When electrical power is supplied to the test set and the alternating current (ac) and direct current (dc) power switches are in the on position as the reset key is pressed, the test set shall:

a. Reset itself.
b. Initialize all input/output parts.
c. Test the central processor.
d. Test the program memory to ensure that the program is correct.
e. Test the random access memory.

The completion of the initialization and self test shall be indicated by the illumination of the pass indicator. The reference display shall indicate 8888 and the code display shall indicate 88. If the positions of the Read-Only-Memory circuit card assemblies, Part Numbers 11831008 and 11831007 are reversed in the card rack assembly, the initialization and self test shall still be performed whenever power is applied or the reset key is pressed. Completion of the initalization and self test in this configuration shall be indicated by the illumination of the pass indicator. The reference display and code display shall indicate 7777 and 77, respectively.

3.6.1.1.2 Automatic self-test. The test set shall execute an automatic self-test when the following sequence of events occur.

a. The self-test cable, Part Number 11830645, is attached to the test set.
b. The cable code "000" is displayed.
c. The code "AO" has been entered on the keyboard.
d. The run key is depressed.

The automatic self-test shall verify proper operation of the input/output circuitry and the sight simulator.

3.6.1.2 Lamp test. When the lamp test key is pressed, the pass, fail, up, left, center, right and reticle indicators shall be illuminated. The reference display and the code display shall indicate +888.8 and 88, respectively.

3.6.1.3 Power indicators. Whenever electrical power is applied to the test set and the ac or dc power switch is in the on position, the corresponding power indicator shall be illuminated.

3.6.1.4 Keyboard. When any of the keys 1 through 9, A or C are pressed, the key designation shall be indicated on the code display. When the first key is pressed, the designation shall be

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indicated on the left digit of the code display with the right digit blank. When the second key is pressed, the designation shall be indicated in the right digit. Subsequent two digit key entries shall be indicated in the same manner.

3.6.1.5 Input/output interface. When selected by the test set program, each input or output pin shall be capable of providing the input or output signals specified in Table I through VI. All ac signals shall be 400 ± 20 hertz (Hz). An ac signal in phase with the ac input power shall be indicated by O 1. A signal out of phase with the input power by 180 degrees shall be indicated by O 2.

TABLE I. - Connector J1 Input/Output Characteristics

Connector Pin	Function	Input Voltage	Maximum Input Current
J1-A	DC Power Input	24-28.5 Vdc	10A
J2-B	DC Power Return		
J3-C	AC Power Input	108-118 Vac, 400Hz	1A
J1-D	AC Power Return		
J1-E	Cable Shield Ground		

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TABLE II. - Connector J2 Input/Output Characteristics

Connector Pin	Function	Input Voltage	Maximum Input Current	Output Voltage	Maximum Output Current
J2-B	Q Signal Return				
J2-C	Q Signal			0-12 Vac	50 A
J2-D	Elevation Error			0-12 Vac	50 A
J2-E	Elevation Error Return				
J2-G	Vx Cos Azimuth Cos Elevation	0-12 Vac	50 A		
J2-H	Vx Return				
J2-J	Azimuth Error			0-12 Vac	50 A
J2-K	Azimuth Error Return				
J2-N	Vz Sin Elevation	0-12 Vac	50 A		
J2-P	Vz Return				
J2-R	Resolver Power	28±1 Vdc			
J2-S	Power Return				
J2-T	Vy Sin Azimuth Cos Elevation	0-12 Vac	50 A		
J2-U	Vy Return				
J2-W	Airspeed Compensation	0-12 Vac	50 A		
J2-X	Airspeed Return			28 Vdc	0.5 A
J2-Y	Action Signal				
J2-a	Action Power	28 Vdc	0.5 A		
J2-f	Reticle Return			Ground	
J2-m	Shield Ground			Ground	

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TABLE III. Connector J3 Input/Output Characteristics

Connector Pin	Function	Maximum Output Current or Load	Output Voltage Resistance
J3-A	DC Output	40 Kilohm Load	+ 78 ± 4 Vdc
		40 Kilohm Load	- 78 ± 4 Vdc
		40 Kilohm Load	+ 48.5 ± 3.0 Vdc
		40 Kilohm Load	- 48.5 ± 3.0 Vdc
		40 Kilohm Load	+ 7.3 ± 0.3 Vdc
		40 Kilohm Load	- 7.3 ± 0.3 Vdc
		40 Kilohm Load	+ 1.95 ± 0.10 Vdc
		40 Kilohm Load	- 1.95 ± 0.10 Vdc
J3-B	DC Output	40 Kilohm Load	+ 78 ± 4 Vdc
		40 Kilohm Load	- 78 ± 4 Vdc
		40 Kilohm Load	+ 48.5 ± 3.0 Vdc
		40 Kilohm Load	- 48.5 ± 3.0 Vdc
		40 Kilohm Load	+ 7.3 ± 0.3 Vdc
		40 Kilohm Load	- 7.3 ± 0.3 Vdc
		40 Kilohm Load	+ 1.95 ± 0.10 Vdc
		40 Kilohm Load	- 1.95 ± 0.10 Vdc
J3-C	Ground		
J3-D	AC Output	1 mA	7.0 ± 0.5 Vac, O1
J3-E	AC Output	1mA	7.0 ± 0.5 Vac, O1
J3-F	Switched Ground		
J3-G	AC Output	250 mA	108-118 Vac, O1
J3-H	Ground		
J3-J	AC Output	100 mA	26 ± 2 Vac, O1
J3-K	Ground		
J3-L	Switched Ground		
J3-M	DC Output	100 mA	28 ± 2 Vdc
J3-N	DC Output	100 mA	28 ± 2 Vdc
J3-P	DC Output	100 mA	28 ± 2 Vdc

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TABLE III. Connector J3 Input/Output Characteristics - Continued

Connector Pin	Function	Maximum Output Current or Load	Output Voltage Resistance
J3-R	DC Output	100 mA	28 ± 2 Vdc
J3-S	DC Output	100 mA	28 ± 2 Vdc
J3-T	DC Output	100 mA	28 ± 2 Vdc
J3-U	DC Output	100 mA	28 ± 2 Vdc
J3-V	DC Output	100 mA	28 ± 2 Vdc
J3-W	DC Output	100 mA	28 ± 2 Vdc
J3-X	DC Output	100 mA	28 ± 2 Vdc
J3-Y	DC Output	100 mA	28 ± 2 Vdc
J3-Z	DC Output	100 mA	28 ± 2 Vdc
J3-a	AC Output	1 mA	5.0 ± 0.2 Vac, O2
		1 mA	7.0 ± 0.5 Vac, O2
J3-b	DC Output	1 mA	0.5 ± 0.1 Vdc
		1 mA	- 0.5 ± 0.1 Vdc
J3-c	Ground		
J3-d	DC Output	1 mA	0.5 ± 0.1 Vdc
		1 mA	- 0.5 ± 0.1 Vdc
J3-e	AC Output	1 mA	2.0 ± 0.2 Vac, O1
		1 mA	2.0 ± 0.2 Vac, O2
		1 mA	4.0 ± 0.4 Vac, O1
		1 mA	4.0 ± 0.4 Vac, O2
J3-f	AC Output	1 mA	8.0 ± 0.5 Vac, O1
		1 mA	0.7 ± 0.1 Vac, O1
		1 mA	2.5 ± 0.2 Vac, O2
J3-g	Switched Ground		
J3-h	Switched Ground		
J3-i	DC Output	100 mA	28 ± 2 Vdc
J3-j	Ground		

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TABLE III. Connector J3 Input/Output Characteristics - Continued

Connector Pin	Function	Maximum Output Current or Load	Output Voltage Resistance
J3-k	DC Output	100 mA	28 ± 2 Vdc
J3-m	DC Output	100 mA	28 ± 2 Vdc
J3-n	DC Output	100 mA	28 ± 2 Vdc
J3-p	DC Output	100 mA	28 ± 2 Vdc
J3-q	DC Output	100 mA	28 ± 2 Vdc
J3-r	DC Output	100 mA	28 ± 2 Vdc
J3-s	DC Output	100 mA	28 ± 2 Vdc
J3-t	DC Output	100 mA	28 ± 2 Vdc
J3-u	Ground		
J3-v	AC Output	1 mA	0.5 ± 0.1 Vac, O1
		1 mA	1.3 ± 0.2 Vac, O1
		1 mA	2.0 ± 0.2 Vac, O1
		1 mA	2.5 ± 0.2 Vac, O1
		1 mA	0.5 ± 0.1 Vac, O2
		1 mA	1.3 ± 0.2 Vac, O2
		1 mA	2.0 ± 0.2 Vac, O2
		1 mA	2.5 ± 0.2 Vac, O2
J3-w	AC Output	1 mA	0.5 ± 0.1 Vac, O1
		1 mA	1.3 ± 0.2 Vac, O1
		1 mA	2.0 ± 0.2 Vac, O1
		1 mA	2.5 ±0.2 Vac, O1
		1 mA	0.5 ± 0.1 Vac, O2
		1 mA	1.3 ± 0.2 Vac, O2
		1 mA	2.0 ± 0.2 Vac, O2
		1 mA	2.5 ± 0.2 Vac, O2
J3-x	AC Output	1 mA	2.5 ± 0.2 Vac, O1

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TABLE III. Connector J3 Input/Output Characteristics - Continued

Connector Pin	Function	Maximum Output Current or Load	Output Voltage Resistance
J3-y	AC Output	1 mA	2.5 ± 0.2 Vac, O2
J3-z	AC Output	1 mA	2.5 ± 0.2 Vac, O2
J3-AA	DC Output	100 mA	28 ± Vdc
J3-BB	DC Output	100 mA	28 ± 2 Vdc
J3-CC	DC Output	100 mA	28 ± 2 Vdc
J3-DD	DC Output	100mA	28 ± 2 Vdc
J3-EE	DC Output	100 mA	28 ± Vdc
J3-FF	Ground		
J3-GG	Switched Ground		
J3-HH	AC Output	1 mA	7.0 ± 0.5 Vac, O2
J3-JJ	Ground		
J3-KK	AC Output	1 mA	7.0 ± 0.5 Vac, O2
J3-LL	AC Output	1 mA	7.0 ± 0.5 Vac, O2
J3-MM	AC Output	1 mA	2.5 ± 0.2 Vac, O1
J3-NN	AC Output	1 mA	7.0 ± 0.5 Vac, O2
J3-PP	Ground		

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TABLE IV. Connector J4 Input/Output Characteristics

Connector Pin	Function	Input Voltage	Input Impedance	Output Voltage	Output Impedance
J4-A	Ground				
J4-B	AC Input	0-14 Vac	100 Kilohms		
J4-C	AC Input	0-14 Vac	100 Kilohms		
J4-D	AC Input	0-14 Vac	100 Kilohms		
J4-E	AC Input	0-14 Vac	100 Kilohms		
J4-F	AC Input	0-14 Vac	100 Kilohms		
J4-G	AC Input	0-14 Vac	100 Kilohms		
J4-H	AC Input	0-14 Vac	100 Kilohms		
J4-J	AC Input	0-14 Vac	100 Kilohms		
J4-K	AC Input	0-14 Vac	100 Kilohms		
J4-L	AC Input	0-14 Vac	100 Kilohms		
J4-M	AC Input	0-14 Vac	100 Kilohms		
J4-N	Ground				
J4-P	Ground				
J4-R	Ground				
J4-S	Cable Code Common			Ground	
J4-T	Cable Code Bit 0			5±1Vdc	5.1 Kilohms
J4-U	Cable Code Bit 1			5±1Vdc	5.1 Kilohms
J4-V	Cable Code Bit 2			5±1Vdc	5.1 Kilohms
J4-W	Cable Code Bit 3			5±1Vdc	5.1 Kilohms

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TABLE IV. Connector J4 Input/Output Characteristics - Continued

Connector Pin	Function	Input Voltage	Input Impedance	Output Voltage	Output Impedance
J4-X	AC Input	0-28 Vac	100 Kilohms		
J4-Y	AC Input	0-115 Vac	320 Kilohms		
J4-Z	AC Input	0-115 Vac	320 Kilohms		
J4-a	AC Input	0-115 Vac	320 Kilohms		
J4-b	DC Input	0-400 Vdc	2 Megohms		
J4-c	DC Input	0-400 Vdc	2 Megohms		
J4-d	DC Input	0-20 Vdc		100 Kilohms	
J4-e	DC Input	0-20 Vdc		100 Kilohms	
J4-f	DC Input	0-20 Vdc		100 Kilohms	
J4-g	DC Input	0-20 Vdc		100 Kilohms	
J4-h	DC Input	0-20 Vdc		100 Kilohms	
J4-i	DC Input	0-30 Vdc		100 Kilohms	
J4-j	DC Input	0-30 Vdc		100 Kilohms	
J4-k	DC Input	0-30 Vdc		100 Kilohms	
J4-m	DC Input	0-30 Vdc		100 Kilohms	
J4-n	DC Input	0-30 Vdc		100 Kilohms	
J4-p	DC Input	0-30 Vdc		100 Kilohms	
J4-q	DC Input	0-30 Vdc		100 Kilohms	
J4-r	DC Input	0-30 Vdc		100 Kilohms	
J4-s	DC Input	0-30 Vdc		100 Kilohms	
J4-t	DC Input	0-30 Vdc		100 Kilohms	
J4-u	DC Input	0-30 Vdc		100 Kilohms	
J4-v	DC Input	0-30 Vdc		100 Kilohms	
J4-w	DC Input	0-30 Vdc		100 Kilohms	
J4-x	DC Input	0-30 Vdc		100 Kilohms	
J4-y	DC Input	0-30 Vdc		100 Kilohms	
J4-z	DC Input	0-30 Vdc		100 Kilohms	

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TABLE IV. Connector J4 Input/Output Characteristics - Continued

Connector Pin	Function	Input Voltage	Input Impedance	Output Voltage	Output Impedance
J4-AA	DC Input	0-30 Vdc		100 Kilohms	
J4-BB	DC Input	0-30 Vdc		100 Kilohms	
J4-CC	DC Input	0-30 Vdc		100 Kilohms	
J4-DD	DC Input	0-30 Vdc		100 Kilohms	
J4-EE	DC Input	0-30 Vdc		100 Kilohms	
J4-FF	DC Input	0-30 Vdc		100 Kilohms	
J4-GG	DC Input	0-30 Vdc		100 Kilohms	
J4-HH	DC Input	0-30 Vdc		100 Kilohms	

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TABLE V. Connector J5 Input/Output Characteristics

Connector Pin	Function	Input Voltage	Input Impedance	Output Voltage	Output Impedance
J5-A	DC Output			28±2 Vdc	100 mA
J5-B	DC Output			28±2 Vdc	100 mA
J5-C	Ground				
J5-D	DC Output			28±2 Vdc	100 mA
J5-E	Cable Code Bit 0			5±1 Vdc	5.1 Kilohms
J5-F	Cable Code Bit 1			5±1 Vdc	5.1 Kilohms
J5-G	DC Input	0-20 Vdc	100 Kilohms		
J5-H	DC Input	0-20 Vdc	100 Kilohms		
J5-J	DC Input	0-20 Vdc	100 Kilohms		
J5-K	Ground				
J5-L	AC Input	0-14 Vac	100 Kilohms		
J5-M	Ground				
J5-N	AC Input	0-14 Vac	100 Kilohms		
J5-P	Ground				
J5-R	AC Input	0-14 Vac	100 Kilohms		
J5-S	Ground				
J5-T	AC Input	0-14 Vac	100 Kilohms		
J5-U	Ground				
J5-V	DC Output			28±2 Vdc	100 mA
J5-W	DC Output			28±2 Vdc	100 mA
J5-X	DC Output			+15±0.5Vdc	10 Kilohms

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TABLE V. Connector J5 Input/Output Characteristics - Continued

Connector Pin	Function	Input Voltage	Input Impedance	Output Voltage	Output Impedance
J5-Y	DC Output			-15±0.5Vdc	10 Kilohms
J5-Z	DC Input	0-30 Vdc	100 Kilohms		
J5-a	Cable Code Bit 2			5±1 Vdc	5.1 Kilohms
J5-b	Cable Code Bit 3			5±1 Vdc	5.1 Kilohms
J5-c	DC Output			5±1 Vdc	10 Kilohms
J5-d	DC Input	0-30 Vdc	100 Kilohms		
J5-e	DC Input	0-30 Dvc	100 Kilohms		
J5-f	Ground				
J5-g	Ground				
J5-h	Ground				
J5-i	Ground				
J5-j	Ground				
J5-k	AC Input	0-28 Vac	100 Kilohms		
J5-m	AC Output			26±2 Vac,01	
J5-n	DC Input	0-20 Vdc	100 Kilohms		
J5-p	DC Input	0-30 Vdc	100 Kilohms		
J5-q	DC Input	0-30 Vdc	100 Kilohms		
J5-r	DC Input	0-30 Vdc	100 Kilohms		
J5-s	DC Input	0-30 Vdc	100 Kilohms		
J5-t	DC Input	0-30 Vdc	100 Kilohms		
J5-u	DC Input	0-30 Vdc	100 Kilohms		
J5-v	DC Input	0-30 Vdc	100 Kilohms		
J5-w	AC Input	0-28 Vdc	100 Kilohms		

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TABLE V. Connector J5 Input/Output Characteristics - Continued

Connector Pin	Function	Input Voltage	Input Impedance	Output Voltage	Output Impedance
J5-x	DC Input	0-30 Vdc	100 Kilohms		
J5-y	DC Input	0-30 Vdc	100 Kilohms		
J5-z	DC Input	0-30 Vdc	100 Kilohms		
J5-AA	DC Input	0-30 Vdc	100 Kilohms		
J5-BB	DC Input	0-30 Vdc	100 Kilohms		
J5-CC	DC Input	0-30 Vdc	100 Kilohms		
J5-DD	DC Input	0-30 Vdc	100 Kilohms		
J5-EE	DC Input	0-30 Vdc	100 Kilohms		
J5-FF	DC Input	0-30 Vdc	100 Kilohms		
J5-GG	DC Input	0-30 Vdc	100 Kilohms		
J5-HH	DC Input	0-30 Vdc	100 Kilohms		

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TABLE VI. Connector J6 Input/Output Characteristics

Connector Pin	Function	Input Voltage	Input Impedance or Maximum Input Current	Output Voltage	Maximum Output Current
J6-A	Ground				
J6-B	DC Input	0-30 Vdc	100 Kilohms		
J6-C	DC Input	0-30 Vdc	100 Kilohms		
J6-D	DC Input	0-30 Vdc	100 Kilohms		
J6-E	DC Input	0-30 Vdc	100 Kilohms		
J6-F	AC Input	0-28 Vac	100 Kilohms		
J6-G	Ground				
J6-H	DC Input	0-30 Vdc	100 Kilohms		
J6-J	DC Input	0-30 Vdc	100 Kilohms		
J6-K	Right Stow Signal	24-28.5 Vdc	100 mA		
J6-M	Up Stow Signal	24-28.5 Vdc	100 mA		
J6-N	DC Limit Switch Power			24-28.5 Vdc	200 mA
J6-P	Center Stow Signal	24-28.5 Vdc	100 mA		
J6-R	DC Limit Switch Power			24-28.5 Vdc	200 mA
J6-S	Left Stow Signal	24-28.5 Vdc	100 mA		
J6-T	DC Input	0-30 Vdc	100 Kilohms		
J6-U	DC Input	0-30 Vdc	100 Kilohms		

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3.6.1.6 Sight simulation. The test set shall include the capability of simulating the AH-1S helicopter sight resolvers and shall be aligned using the procedure defined in Technical Manual TM 9-4933-224-13 & P.

3.6.1.7 Firing simulator. The firing simulator shall produce a 20 + 20 Vdc level on the fire volts power supply output for a duration of 1 to 15 milliseconds. This waveform shall be repetitive at a rate of 8 to 22 Hz.

3.6.1.8 Cable codes. After completion of initialization and self test, when the run key is pressed, the test set shall display in the reference display a four-digit code indicating which test cable is attached. The code shall indicate which unit is to be tested and the test set program will select the proper tests. The code for each unit is specified in Table VII. If no cable is attached, the reference display shall indicate 9991.

TABLE VII. - Cable Codes

Cable Code	Unit Under Test	Cable	Part No.
000	Test Set Self Test	W13	11830645
1000	Turret System	W6	11830638
2000	Interface Control Unit - M97E1	W7	11830639
+2000	Interface Control Unit - M97E2	W20	11830952
3000	Turret Control Unit	W8	11830640
4000	Gun Control Unit	W9	11830641
5000	Logic Control Unit	W14	11830646
6000	Emergency Stow Control Unit	W10	11830642
7000	Gunner Control Panel - M97E1	W11	11830643
+7000	Gunner Control Panel - M97E2	W22	11831053
8000	Pilot Control Panel - M97E2	W12	11830644
+8000	Pilot Control Panel - M97E2	W23	11831052

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3.6.1.9 Key codes. If an unallowable key code is entered on the keyboard and the run key depressed, the code display shall flash until a new code is entered. The key codes for each unit are specified in Table VIII. All the digital voltmeter codes defined in 3.2.1.10.3 shall be allowable for use on the test set.

TABLE VIII. - Key Codes

Unit Under Test	Key Codes
Test Set Self Test	AO, CO, C1, C2, C3, C4 C5, C6, C7, C8, C9
Interface Control Unit - M97E1	AO, CO, C1, C2, C3, C4 C5, C6
Interface Control Unit - M97E2	AO, CO, C1, C2, C3, C4 C5
Turret Control Unit	AO, CO, C1, C2
Gun Control Unit	AO, CO, C1
Logic Control Unit	AO, A1, CO, C1, C2, C3 C4, C5
Emergency Stow Control Unit	AO
Gunner Control Panel - M97E1	AO, A1, A2, CO, C1
Gunner Control Panel - M97E2	AO, A1, A2, CO, C1
Pilot Control Panel - M97E1	AO, A1, A2, A3, CO, C1
Pilot Control Panel - M97E2	AO, A1, A2, A3, CO, C1
AVUM System Test	AO, A1, A2, A3, A4, A5, A6, A7, A8, A9

3.6.1.10 Operating modes. The test set shall be capable of operating in the three basic modes specified in the following subparagraphs.

3.6.1.10.1 Automatic test mode. The test set shall execute an automatic test of the unit under test when the following sequence of events occur.

a. A cable listed in Table VII is attached to the test set.

b. The cable code has been displayed.

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c. An allowable two-digit code, the first digit of which is "A", has been entered on the keyboard.

d. The run key is depressed.

Signals shall be supplied to the unit under test and the returned signals shall be measured and compared with present limits. If all the returned signals are within the limits, the pass indicator shall be illuminated. If any of the returned signals are outside the limits, the fail indicator shall be illuminated and a four-digit code shall be indicated on the reference display to define which measurement has failed. If more than one measurement fails, fault codes representing the other failures shall be displayed by pressing the next line key.

3.6.1.10.2 Manual test mode. The test set shall provide a predetermined set of stimuli to the unit under test when the following sequence of events occur.

a. A cable listed in Table VII is attached to the test set.

b. A cable code has been displayed.

c. An allowable two-digit code, the first digit of which is "C" has been entered on the keyboard.

d. The run key is depressed.

When the test is complete, the reference display shall indicate 9993. The test stimuli shall remain on until an automatic test or a new manual test is run, or the test set is reset.

3.6.1.10.3 Digital voltmeter mode. The reference display shall indicate the voltage present on an operator selector input pin, when the following sequence of events occur.

a. A cable listed in Table VII is attached to the test.

b. The cable code has been displayed.

c. A two-digit numerical code is entered through the keyboard.

d. The run key is depressed.

The digital voltmeter input codes and the connector pins which will be selected are specified in Table IX.

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TABLE IX. Digital Voltmeter Codes

Code	Connector Pin	Code	Connector Pin
01	J4-d	41	J4-CC
02	J4-e	42	J4-DD
03	J4-f	43	J4-EE
04	J4-g	44	J4-FF
05	J4-h	45	J4-GG
06	J4-B	46	J4-HH
07	J4-C	47	J5-G
08	J4-D	48	J5-H
09	J4-E	49	J5-n
10	J4-F	50	J5-p
11	J4-G	51	J5-k, J6-F
12	J4-H	52	J5-W
13	J4-J	53	J5-J
14	J4-K	54	J5-L
15	J4-L	55	J5-N
16	J4-M	56	J5-R
17	J4-m	57	J5-T
18	J4-n	58	J5-Z, J6-H, J6-J
19	J4-b	59	J5-d, J6-C
20	J4-c	60	J5-q, J6-U

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TABLE IX. Digital Voltmeter Codes - Continued

Code	Connector Pin	Code	Connector Pin
21	J4-X	61	J5-r
22	J4-Y	62	J5-s, J6-E
23	J4-Z	63	J5-t
24	J4-a	64	J5-u
25	J4-i	65	J5-v
26	J4-j	66	J5-x
27	J4-k	67	J5-y, J6-D
28	J4-p	68	J5-z
29	J4-q	69	J5-AA
30	J4-r	70	J5-BB
31	J4-s	71	J5-CC, J6-B
32	J4-t	72	J5-DD
33	J4-u	73	J5-EE
34	J4-v	74	J5-FF
35	J4-w	75	J5-GG, J6-T
36	J4-x	76	J5-HH
37	J4-y	77	J5-e
38	J4-z		
39	J4-AA		
40	J4-BB		

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3.6.1.11 Proof test (first article inspection only). Test Set Units shall operate when assembled to an accepted AH-1S helicopter.

3.7 Design and construction.

3.7.1 Production drawings. The test set shall be fabricated and assembled in accordance with the drawings, parts tests, and other documents listed on Drawing 11830908.

3.7.2 Standards of manufacture. The test set shall be manufactured to meet the requirements of the applicable drawings.

3.7.3 Workmanship. Workmanship for mechanical and electronic assemblies shall be in accordance with MIL-T-28800. Finished items and parts shall be fabricated in a thorough, workmanlike manner, and, all manufacturing, processing, and assembly operations shall be correctly performed. The parts shall not exhibit faulty material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, burrs, deformations, unblended radii, chips, dents, rust, dirt, grease and oil (except where specifically required), corrosion products, flux, other foreign matter, and missing operations which may affect serviceability, operation, or safety. The cleaning method used shall not be injurious to any part or assembly nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required markings and stamping shall be neat and sharply defined. Required packaging shall be dry. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing the wire insulation. Ends of wires shall be free from burrs, sharp edges or points.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-W-63150 shall apply and are hereby made a part of this detail specification.

4.2 Certification of provisions. The certification provisions of MIL-W-63150 shall apply.

4.3 Classification of inspection. The following types of inspection shall be conducted on this item.

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a. First Article Inspection.
b. Quality Conformance Inspection.

4.4 First article inspection.

4.4.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Test Set Electronic Systems, M137	11831041	5
Case Assembly	11831046	5
Firing Simulator Assembly	11830648	5
Case, Cable	11830714	5
Test Set Assembly	11830999	5
Pannel Assembly Front	11831072	5
Case Assembly	11831001	5
Circuit Card Assembly, Programmable Memory (A)	11831008	5
Circuit Card Assembly, Programmable Memory (B)	11831007	5

4.4.2 Inspections to be performed. First article assemblies components and test specimens may be subjected by the Government to all of the examinations and tests specified in the item detail specification and to any requirements of the applicable drawings. See Table X specified herein.

4.4.3 Rejection. If any assembly, component, or test specimen fails to comply with any applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample that fails to comply with any of the stated requirements.

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TABLE X. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-T-70431(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
	Test Set, Electronic System		See Below
			NEXT HIGHER ASSEMBLY ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Test Set, Universal Turret Subsystem (Dwg 11831041)				
	Examination for defects	5		3.3	4.5.2.9
	Interchangeability	3		3.4	4.6.3
	Proof test	3		3.6.11	4.6.4
	Workmanship	3		3.7.3	Visual
	Case assembly, cable. (Dwg 11831046)				
	Examination for defects	5		3.3	4.5.2.1
	Firing simulator assembly. (Dwg 11830648)				
	Examination for defects	5		3.3	4.5.2.2
	Case, cable. (Dwg 11830714)				
	Examination for defects	5		3.3	4.5.2.3
	Test set assembly. (Dwg 11830999)	5		3.3	4.5.2.4

NOTES:

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TABLE X First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-T-70431(AR)

PARAGRAPH CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
	Test set, electronic system		SHEET 2 OF 2		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY: --- PARAGRAPH REFERENCE /INSPECTION METHOD
	Panel assembly front. (Dwg 11831072)				
	Examination for defects	5		3.3	4.5.2.5
	Case assembly. (Dwg 11831001)				
	Examination for defects	5		3.3	4.5.2.6
	Circuit card programmable memory (A) (Dwg 11831008)				
	Examination for defects	5		3.3	4.5.2.7
	Circuit card programmable memory (B) (Dwg 11831007)				
	Examination for defects	5		3.3	4.5.2.8

NOTES:

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4.5 Quality conformance (acceptance) inspection. Acceptance inspection shall be performed and shall be in accordance with the following:

4.5.1 Lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with the requirements.

4.5.2 Examination. Examination for the inspection listed in the following Classification of Defects and Tests paragraphs shall be performed by sampling to the AQL specified unless 100 percent inspection is indicated. Unless otherwise specified sampling shall be in accordance with MIL-STD-105 Level II, except the contractor may request permission from the procuring activity to use an equivalent continuous sampling plan from MIL-STD-1235.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11831046
4.5.2.1	Case Assembly, Cable				NEXT HIGHER ASSEMBLY 11831041
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Case damaged (puncture, dented or deformed)		0.65%	3.3	Visual
102	Subassemblie(s) damaged		0.65%	3.3	Visual
103	Subassemblie(s) missing		0.65%	3.3	Visual
104	Identification plate improperly located		0.65%	3.3	Visual
Minor					
201	Stencil characters not located as shown		1.5%	3.3	Visual
202	Marking misleading, illegible, missing or ink type incorrect		1.5%	3.3	Visual
203	Size of lettering incorrect		1.5%	3.3	Visual
204	Poor workmanship		2.5%	3.7.3	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-70431(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.2.2	Firing Simulator Assembly	1 OF 1	11830648
			NEXT HIGHER ASSEMBLY 11831046

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder workmanship, unsatisfactory		0.65%	3.3	Visual
102	Wiring workmanship, unsatisfactory		0.65%	3.3	Visual
103	Wiring not the right type		0.65%	3.3	Visual
104	Compound missing		0.65%	3.3	Visual
105	Solder connections not insulated		0.65%	3.3	Visual
106	Part or assembly missing		0.65%	3.3	Visual
107	Part or assembly incorrectly assembled		0.65%	3.3	Visual
108	Hardware, screw or nut missing or loose		0.65%	3.3	Visual/Manual
Minor					
201	Identification plate missing		1.5%	3.3	Visual
202	Marking misleading, illegible, missing or ink type incorrect		1.5%	3.3	Visual
203	Poor workmanship		1.5%	3.7.3	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-70431(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.2.3	Case, Cable	1 OF 1	11830714
			NEXT HIGHER ASSEMBLY 11831046

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pressure relief vacuum, incorrect		0.65%	3.3	Visual
102	Latch assembly and handle plating improper		0.65%	3.3	Visual
103	Material improper		0.65%	3.3	4.6.1
104	Color incorrect				
105	Hardware broken, loose		0.65%	3.3	Visual
106	Closure unsatisfactory		0.65%	3.3	Visual/Manual
107	Case damaged (puncture, dented or deformed)		0.65%	3.3	Visual
108	Bail stops at improper position		0.65%	3.3	Visual
Minor					
201	Length		1.5%	3.3	SMTE
202	Width		1.5%	3.3	SMTE
203	Height		1.5%	3.3	SMTE
204	Marking misleading, illegible or missing		1.5%	3.3	Visual
205	Poor workmanship		2.5%	3.7.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-70431(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.2.4	Test Set Assembly	1 OF 1	11830999
			NEXT HIGHER ASSEMBLY 11831041

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardware, screw or rivet loose or missing		100%	3.3	Visual
102	Washer improper between panel and jumper		100%	3.3	Visual
103	Closure unsatisfactory		100%	3.3	Visual/Manual
104	Part or assembly damaged		100%	3.3	Visual
105	Part or assembly missing		100%	3.3	Visual
106	Part or assembly incorrectly assembled		100%	3.6	Visual
Minor					
201	Identification plate lettering misleading, unidentifiable or missing		1.5%	3.3	Visual
202	Marking improper, missing, illegible or marked with improper ink		1.5%	3.3	Visual
203	Poor workmanhsip		2.5%	3.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-70431(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.2.5	Pannel Assembly Front	1 OF 1	11831072
			NEXT HIGHER ASSEMBLY 11830999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardware, screw, nut or packing missing, or loose		0.65%	3.3	Visual
102	Parts damaged		0.65%	3.3	Visual
103	Part or assembly missing		0.65%	3.3	Visual
104	Part or assembly incorrectly assembled		0.65%	3.3	visual
105	Safety wire twist method improper		0.65%	3.3	Visual
106	Solder connections not insulated		0.65%	3.3	Visual
107	Location of danger identification plate, incorrect		0.65%	3.3	Visual
Minor					
201	Marking misleading, illegible, missing or marked with improper ink		1.5%	3.3	Visual
202	Poor workmanship		2.5%	3.7.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-70431(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 11831001
4.5.2.6	Case Assembly		NEXT HIGHER ASSEMBLY 11830999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder connections not insulated		0.65%	3.3	Visual
102	All exposed previously painted surfaces not touched up with matched paint		0.65%	3.3	Visual
103	Part or assembly missing		0.65%	3.3	Visual
104	Part or assembly damaged		0.65%	3.3	Visual
105	Wire marking improper or missing		0.65%	3.3	Visual
106	Improper wire type		0.65%	3.3	Visual
107	Part or assembly incorrectly assembled		0.65%	3.3	Visual
108	Hardware, screws or rivits loose or missing		0.65%	3.3	Visual/Manual
Minor					
201	Marking, misleading, illegible or missing		1.5%	3.3	Visual
202	Poor workmanship		2.5%	3.7.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.2.7	Circuit Card Assembly Programmable Memory (A)	1 OF 1	11831008
			NEXT HIGHER ASSEMBLY 11831001

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder workmanship unsatisfactory		0.65%	3.3	Visual
102	Wiring workmanship unsatisfactory		0.65%	3.3	Visual
103	Part or assembly missing		0.65%	3.3	Visual
104	Part or assembly damaged		0.65%	3.3	Visual
105	Part or assembly incorrectly assembled		0.65%	3.3	Visual
106	Compoind on areas where not intended		0.65%	3.3	Visual
107	Fasteners loose or missing		0.65%	3.3	Visual/Manual
Minor					
201	Marking misleading, illegible, missing or marked with improper ink		1.5%	3.3	Visual
202	Identification plate missing		1.5%	3.3	Visual
203	Poor workmanship		2.5%	3.7.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-70431(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11831007
4.5.2.8	Circuit Card Assembly Programmable Memory (B)				NEXT HIGHER ASSEMBLY 11831001
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder workmanship unsatisfactory		0.65%	3.3	Visual
102	Wiring workmanship unsatisfactory		0.65%	3.3	Visual
103	Part or assembly missing		0.65%	3.3	Visual
104	Part or assembly damaged		0.65%	3.3	Visual
105	Part or assembly incorrectly assembled		0.65%	3.3	Visual
106	Compound on areas where not intended		0.65%	3.3	Visual
107	Fasteners loose or missing		0.65%	3.3	Visual/Manual
Minor					
201	Marking misleading, illegible, missing or marked with improper ink		1.5%	3.3	Visual
202	Identification plate missing		1.5%	3.3	Visual
203	Poor workmanship		2.5%	3.7.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-T-70431(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.2.9	Test Set, Universal Turret Subsystem	1 OF 1	11831041
			NEXT HIGHER ASSEMBLY ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Performance	a	---	3.6.1	4.6.2
Minor 201	Carrying case damaged, dented or deformed		1.5%	3.3	Visual
202	Poor workmanship		2.5%	3.7.3	Visual

NOTES:

a. Three samples; accept if one fail, reject if one fails

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4.5.5 Testing. The tests (First Article and Quality Conformance) listed herein, shall be performed on inspection lots as defined in 4.5.1 of this specification. The sampling plans shall conform to the provisions of 4.5.2 above.

4.5.6 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection column in the tables, starting with paragrphs 4.5.4.1 and test methods paragraph (see 4.6). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.3 herein.

4.6 Methods of inspection.

4.6.1 Materials, components and processess. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, sub-assemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processess have been followed.

4.6.1.1 Visual and mechanical inspection. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.5.2 and Table X.

4.6.2 Performance test. The Electronic Test Set shall be tested for compliance with specified performance requirements. Each cable assembly included in the test set cable assembly shall be subjected to continuity and dielectric tests to determine that it complies with the specific drawing for the individual assembly. The test is operated within the performance parameters of 3.6.1.1 through 3.2.1.9. In addition, these tests shall ensure that the performance requirements for the operating modes specified in 3.6.1.10 are met. Any M137 Electronic System Test Set which fails to comply with the specified requirement or evidence of degradation of its operation shall be classed defective. Test equipment and procedure shall be satisfactory to the contracting officer (see 4.5.6).

4.6.3 Interchangeability. Any assembly which fails to comply with the specified requirements or evidences degradation of its operational performance shall be satisfactory to the contracting officer (see 4.5.6).

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4.6.4 Proof test (first article inspection only). When specified in the contract (see 6.2) three (3) M137 Electronic System Test Sets shall be forwarded to the site specified by the contracting agency and utilized on an AH-1S helicopter to determine that the test set will function properly within the system. The test sets shall function in accordance to the intent of the design and with no evidence of malfunction.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Test sets shall be preserved, packaged and marked in accordance with the requirements of the Preservation and Packaging Data for Drawing P11831041.

5.1.1 Spare parts or replacement parts. Spare parts or replacement parts shall be cleaned, preserved, packaged as specified in the contract or order.

6. NOTES

6.1 Intended use. The test set assembly covered by this specification is intended for use with the AH-1S helicopter to provide for checkout and fault isolation of the Armament Subsystem, Helicopter, 20MM Automatic Gun: M97E1 and M97E2 and its related components.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements

a. Title, number and date of this specification.
b. Quantity required and delivery schedules.
c. Serialization reuirements, if applicable.
d. Quality Conformance Inspection, if other than spcified in Section 4 of this specification.
e. First Article sample requirements, if other than specified in Section 4 of this specification.
f. Packaging requirements, if other than specified in Section 5 of this specification.
g. Certificates of conformance for each lot or shipment of product.
h. Location to which items are to be shipped.

6.2.2 Contract data requirements. Contract data requirements for inspection equipment design shall conform to Data Item Description DI-R-1714, Tailored.

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6.3 Submission for inspection equipment design approval. See MIL-W-63150. Equipment designs shall be submitted as required to: Commander, AMCCOM, ATTN: DRSMC-QAF-I(D), Dover, NJ 07801. Request letter of submittal shall state contractor, contract number, specification number, item nomenclature, and classification of defects and test paragraphs.

6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to: Commander, AMCCOM, ATTN: DRSMC-QAF-S(D), Dover, NJ 07801.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-A330)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-70431 (AR)	TEST SET, ELECTRONIC SYSTEM: M137

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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